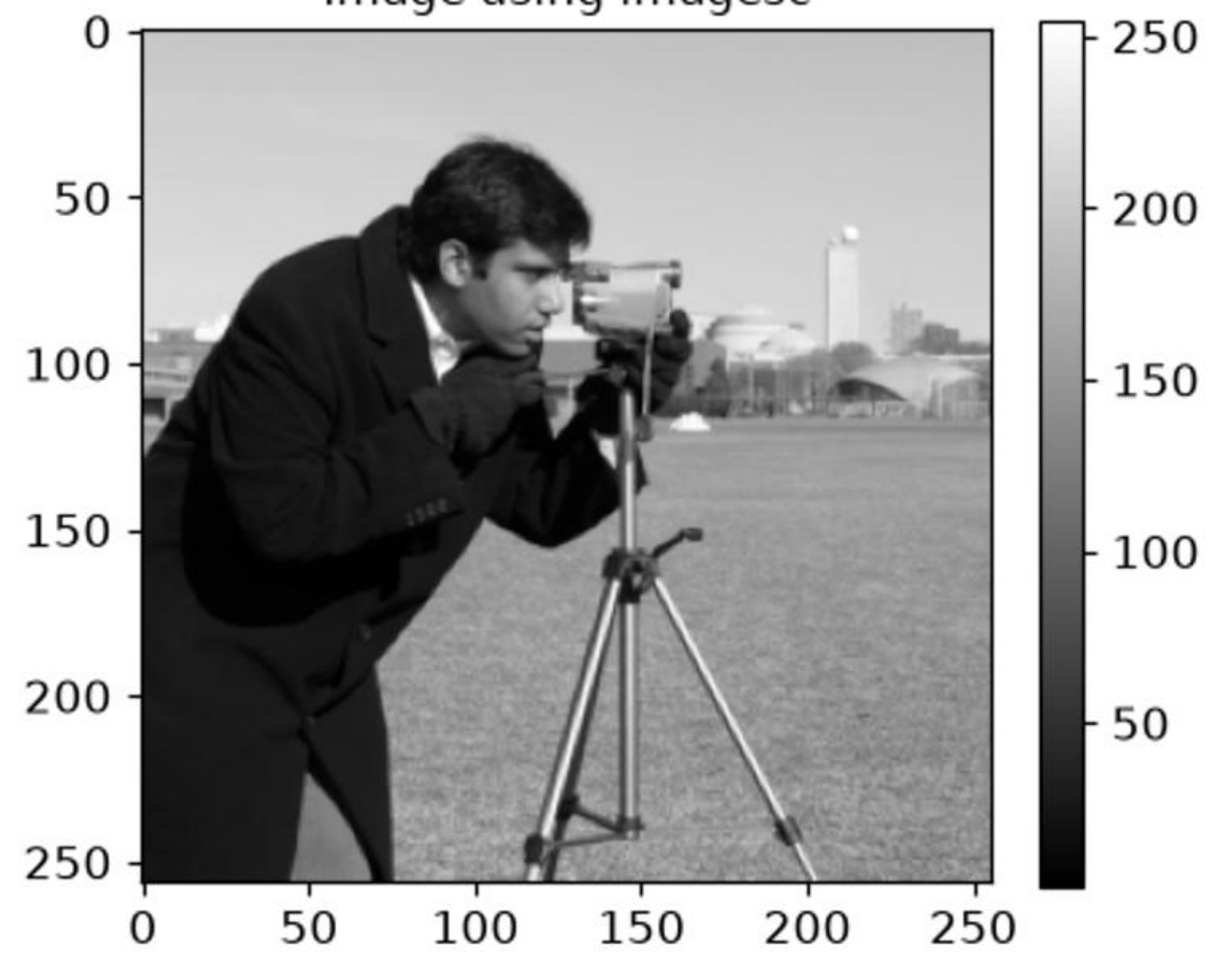


Original Image



Image using imagesc



Experiment 2 — Image Classes, Types and File Formats

RGB Image



Grayscale Image



Binary Image



Indexed Image (64 colors)



Experiment 3 — Arithmetic (top) and Logical (bottom) Operations

Addition



Subtraction



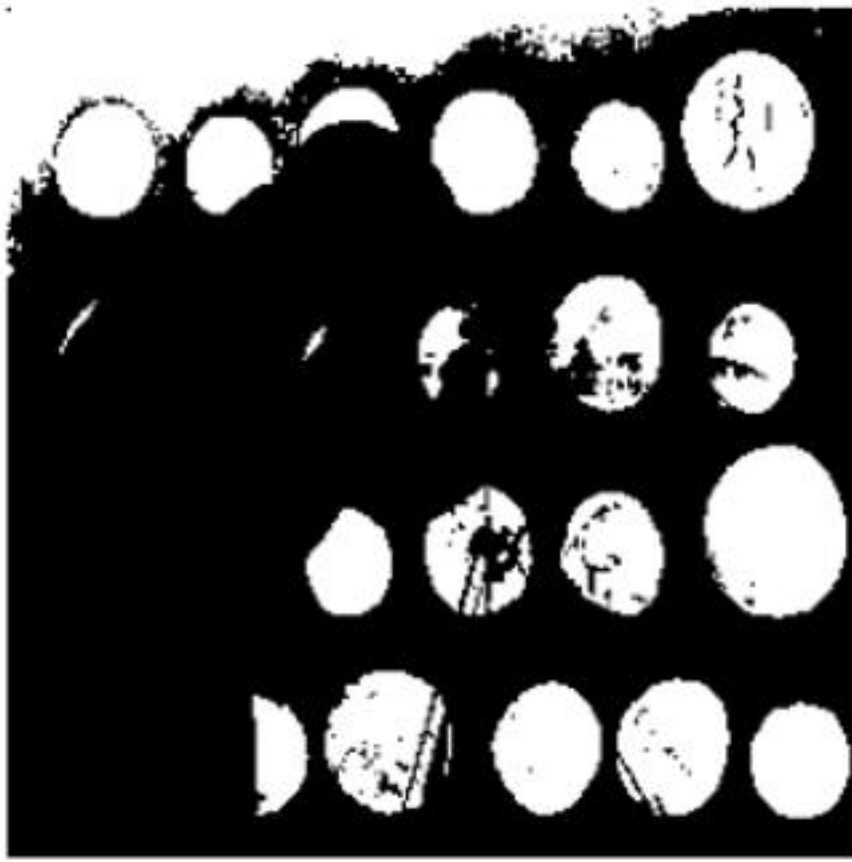
Multiplication



Division



AND



OR



XOR



NOT



Experiment 4 — Zooming, Shrinking and Bit-Plane Slicing

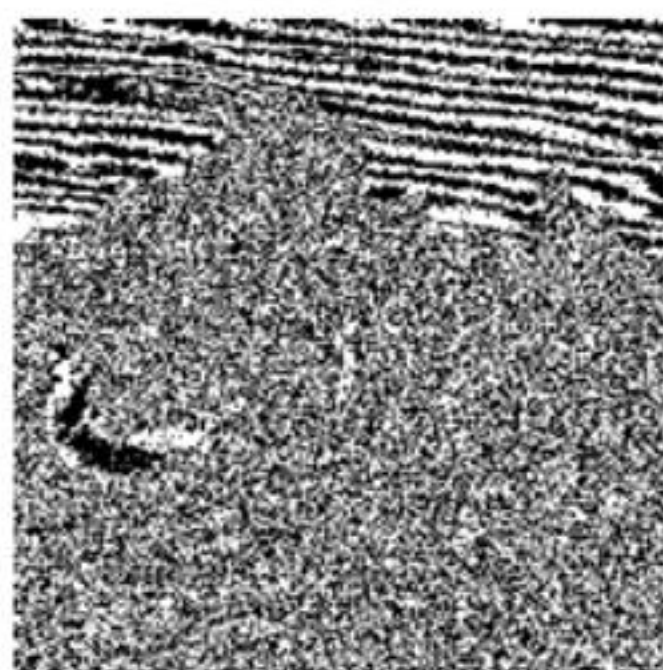
Zoomed Image (2x, bicubic)



Shrunk Image (0.5x, nearest)



Bit Plane 0



Bit Plane 1



Bit Plane 2



Bit Plane 3



Bit Plane 4



Bit Plane 5



Bit Plane 6

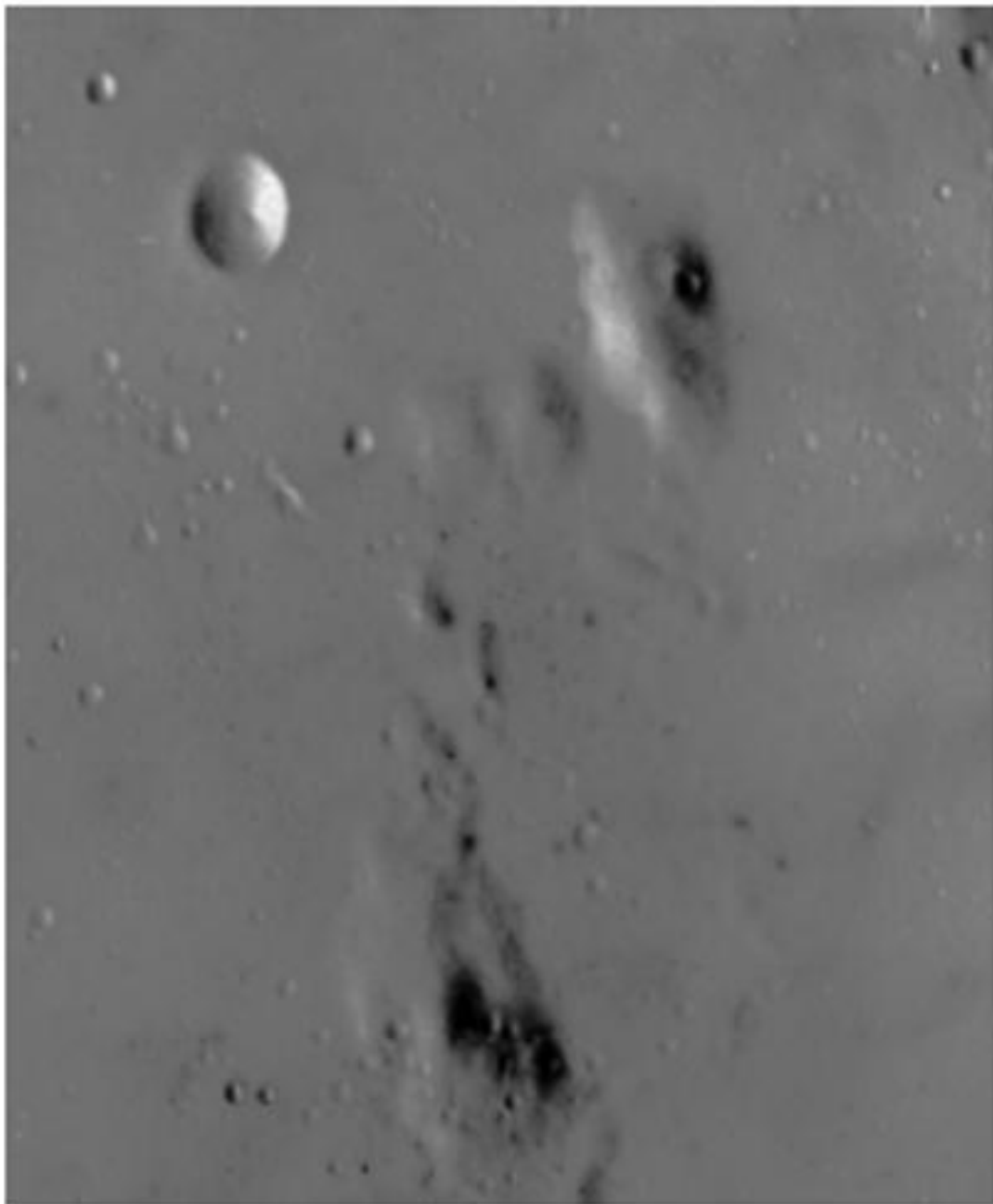


Bit Plane 7

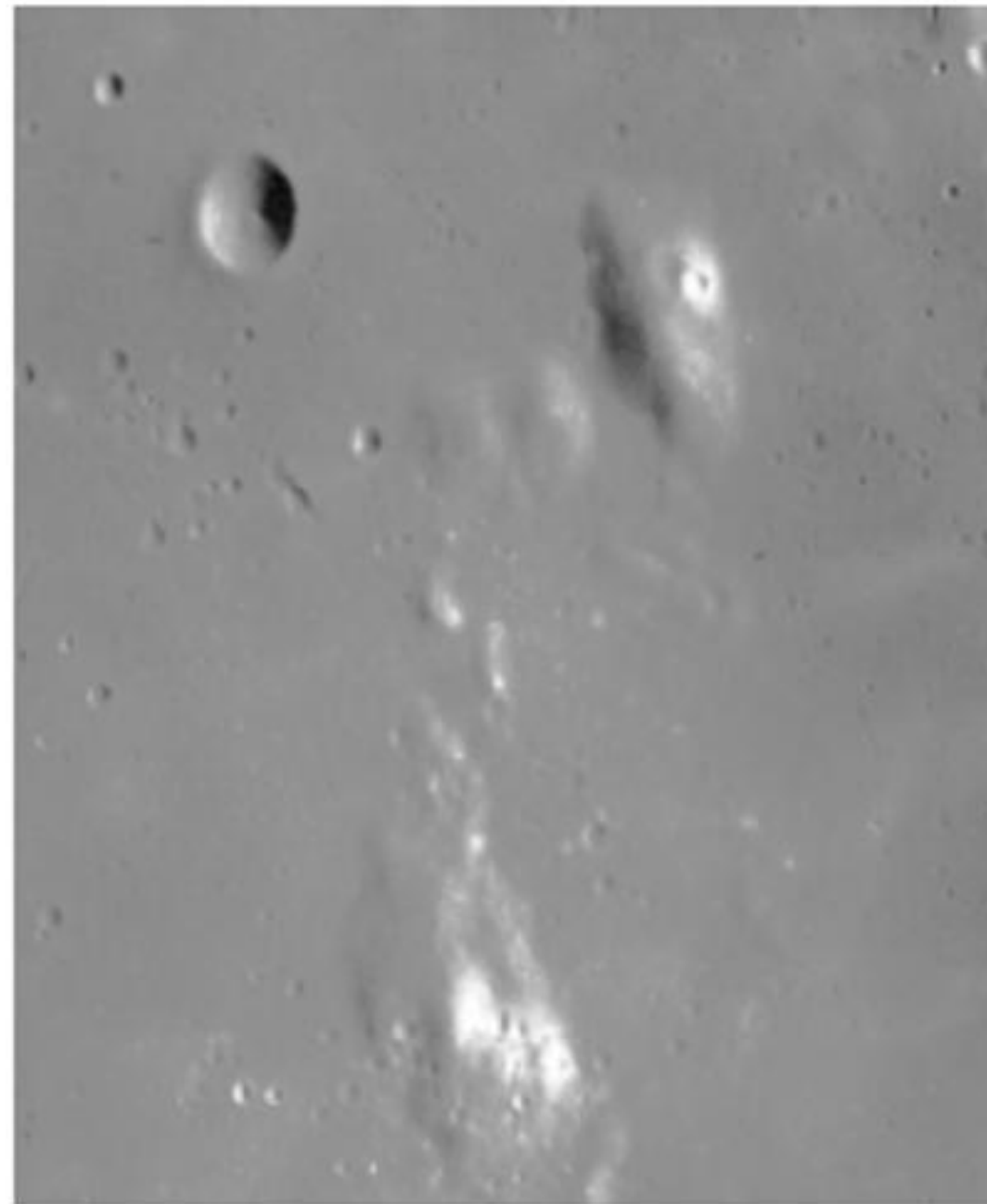


Experiment 5 — Gray-Level Transformations

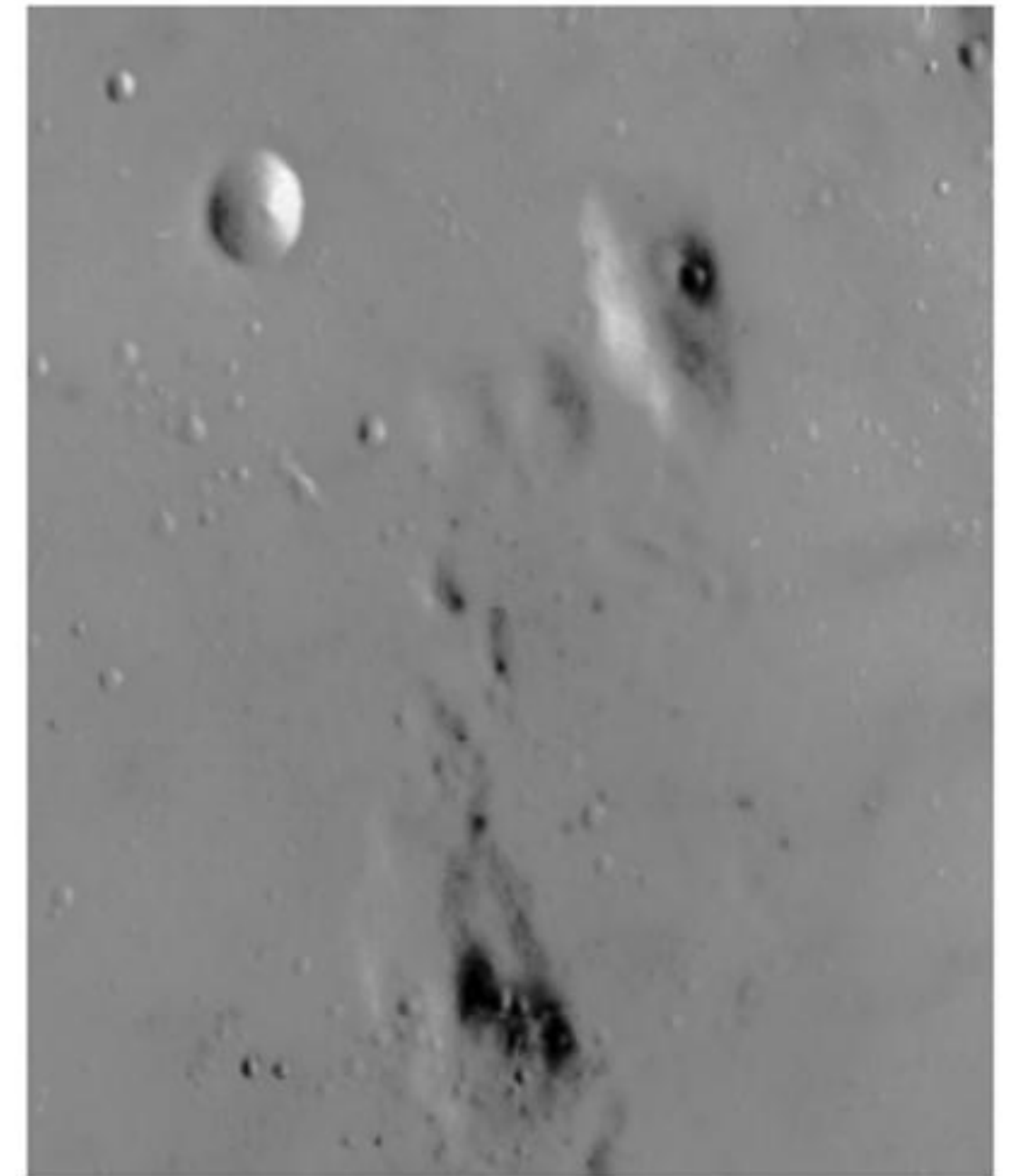
Original



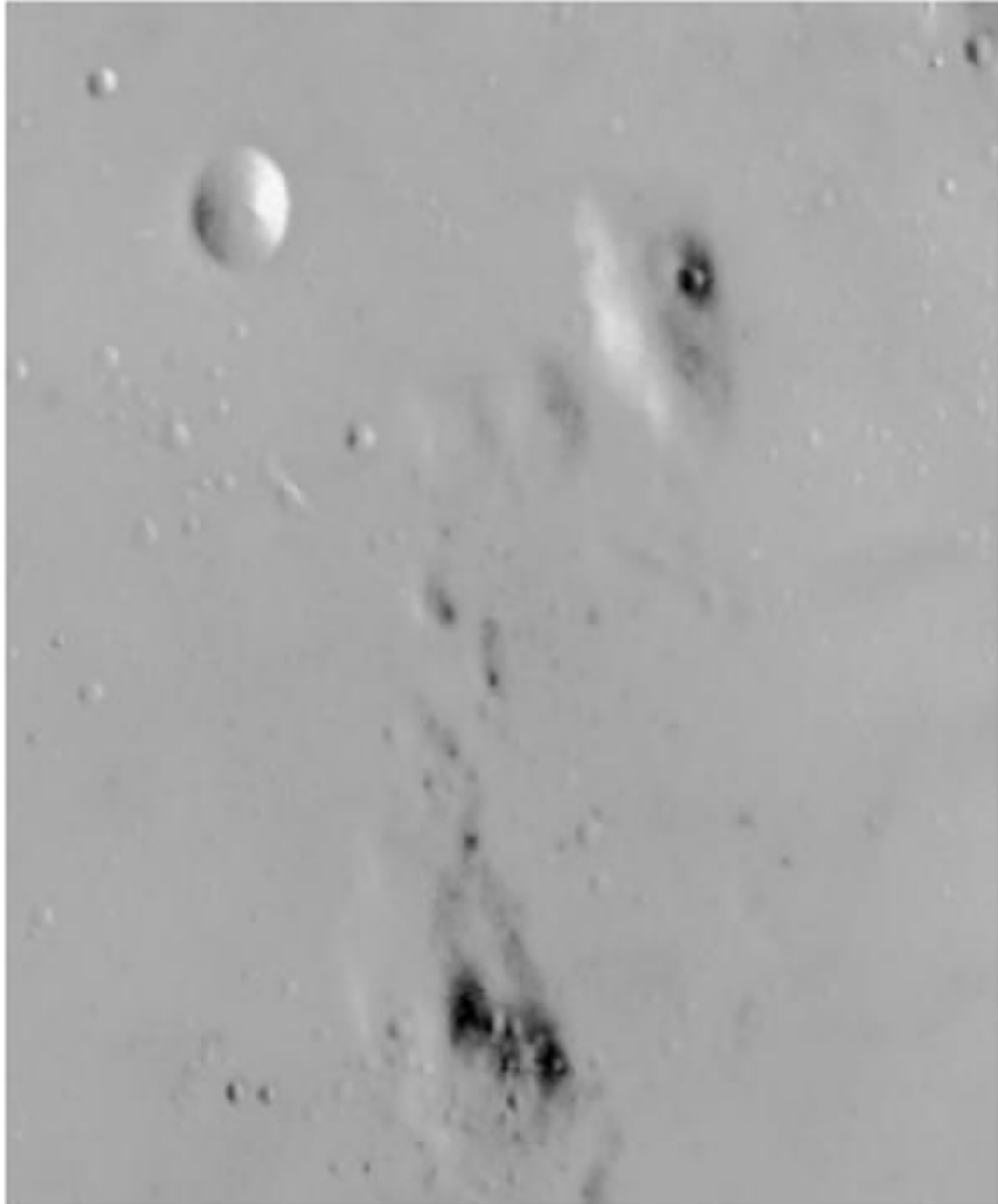
Negative



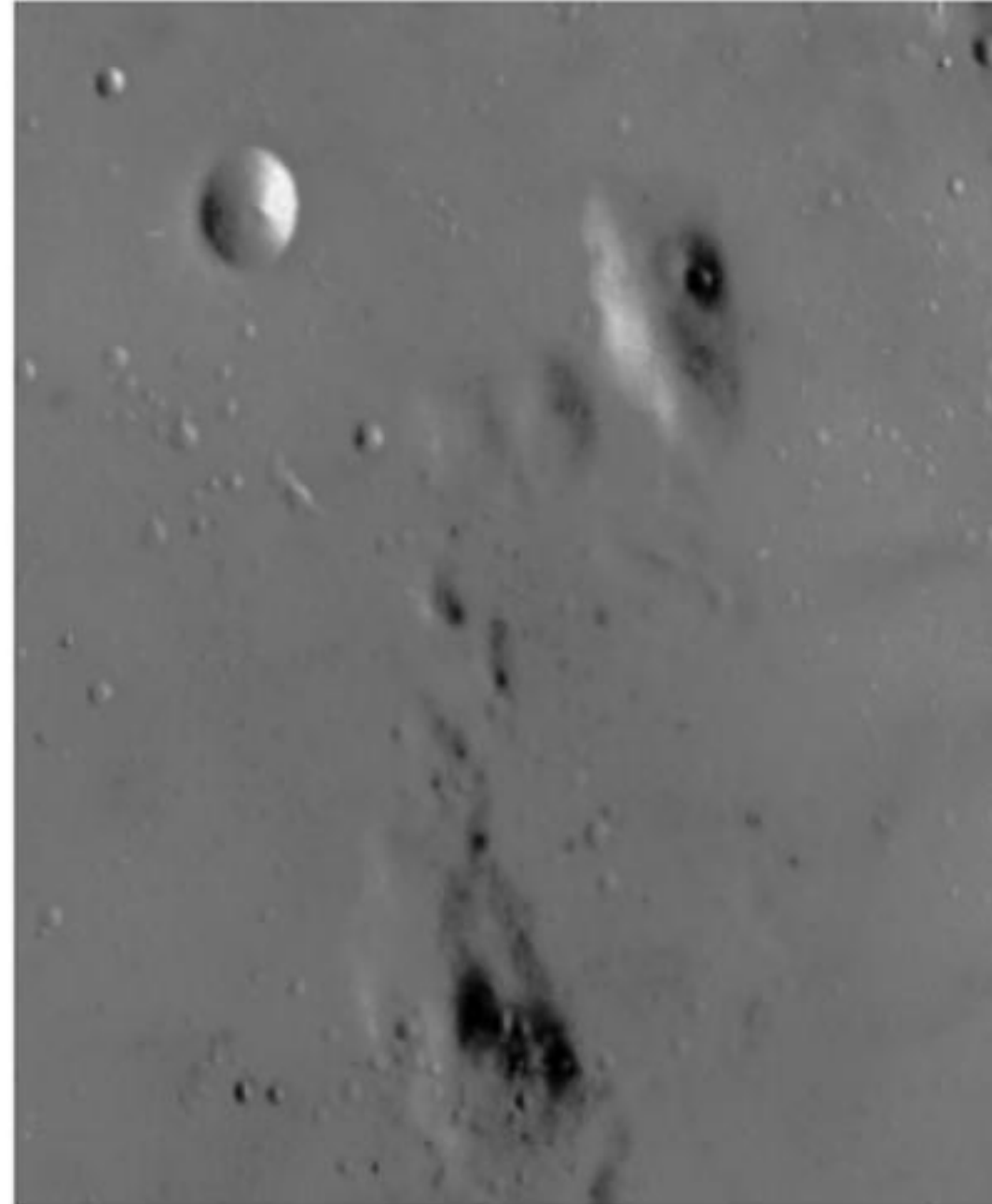
Log Transform



Power-Law (gamma=0.5)

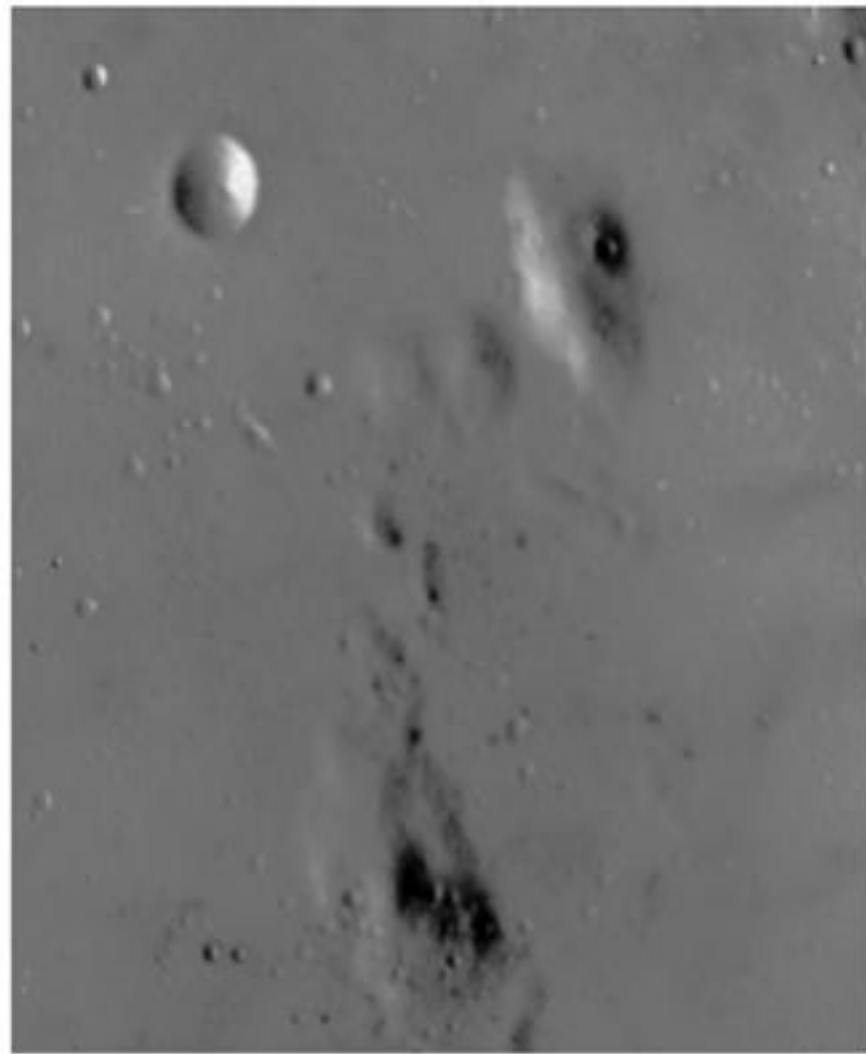


Contrast Stretched

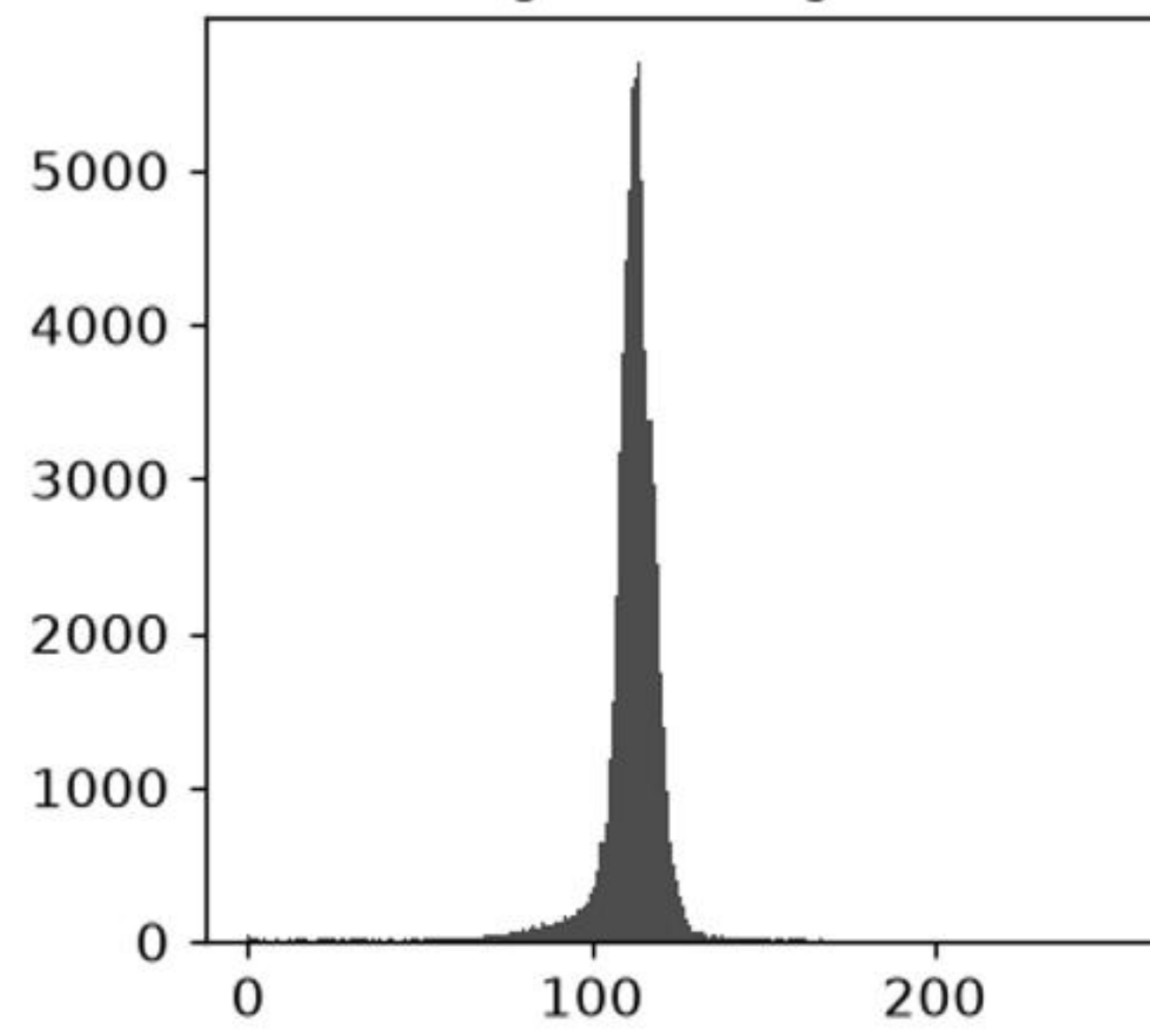


Experiment 6 — Histogram Equalization and Matching

Original (pout)



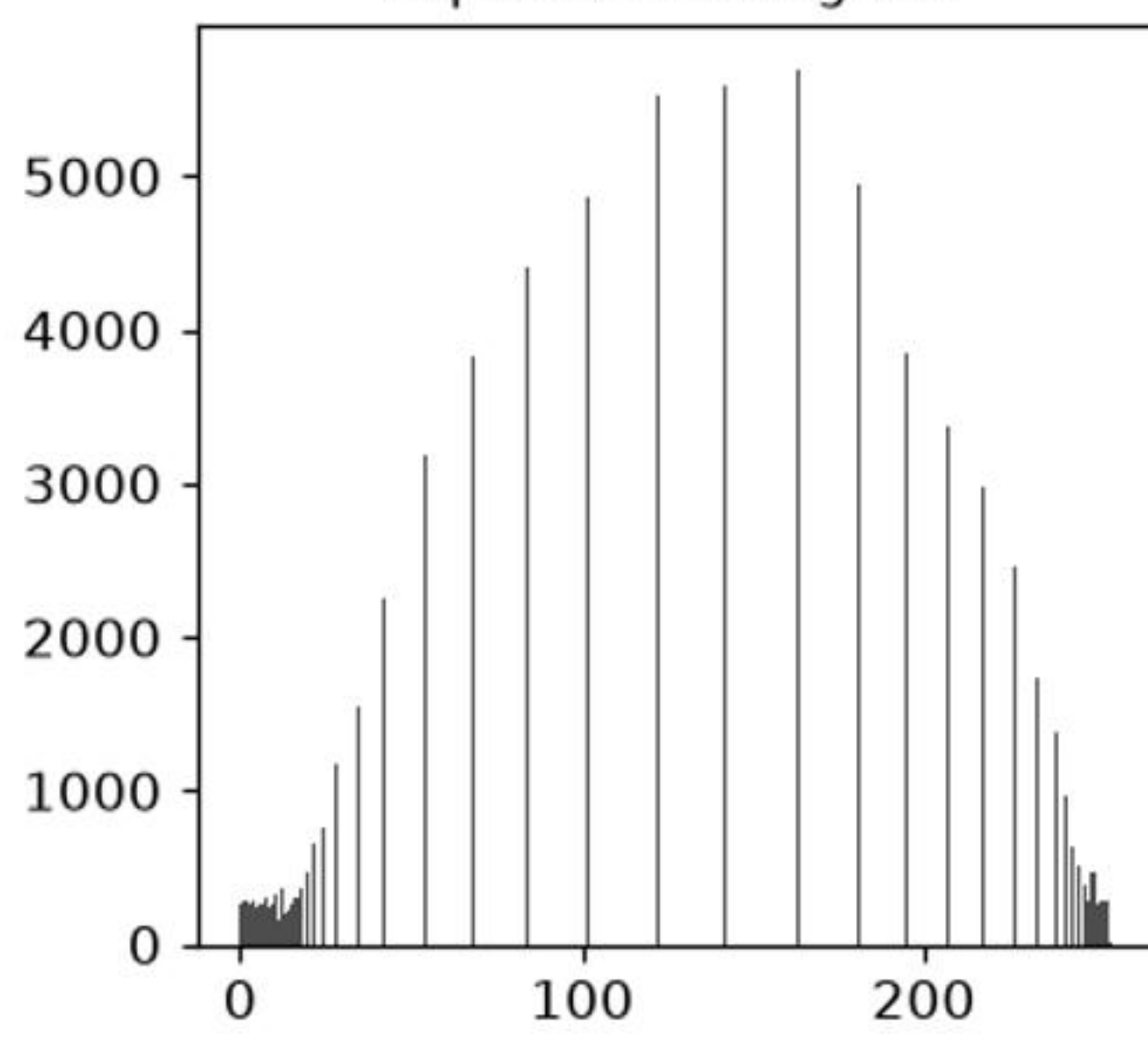
Original Histogram



Histogram Equalized Image



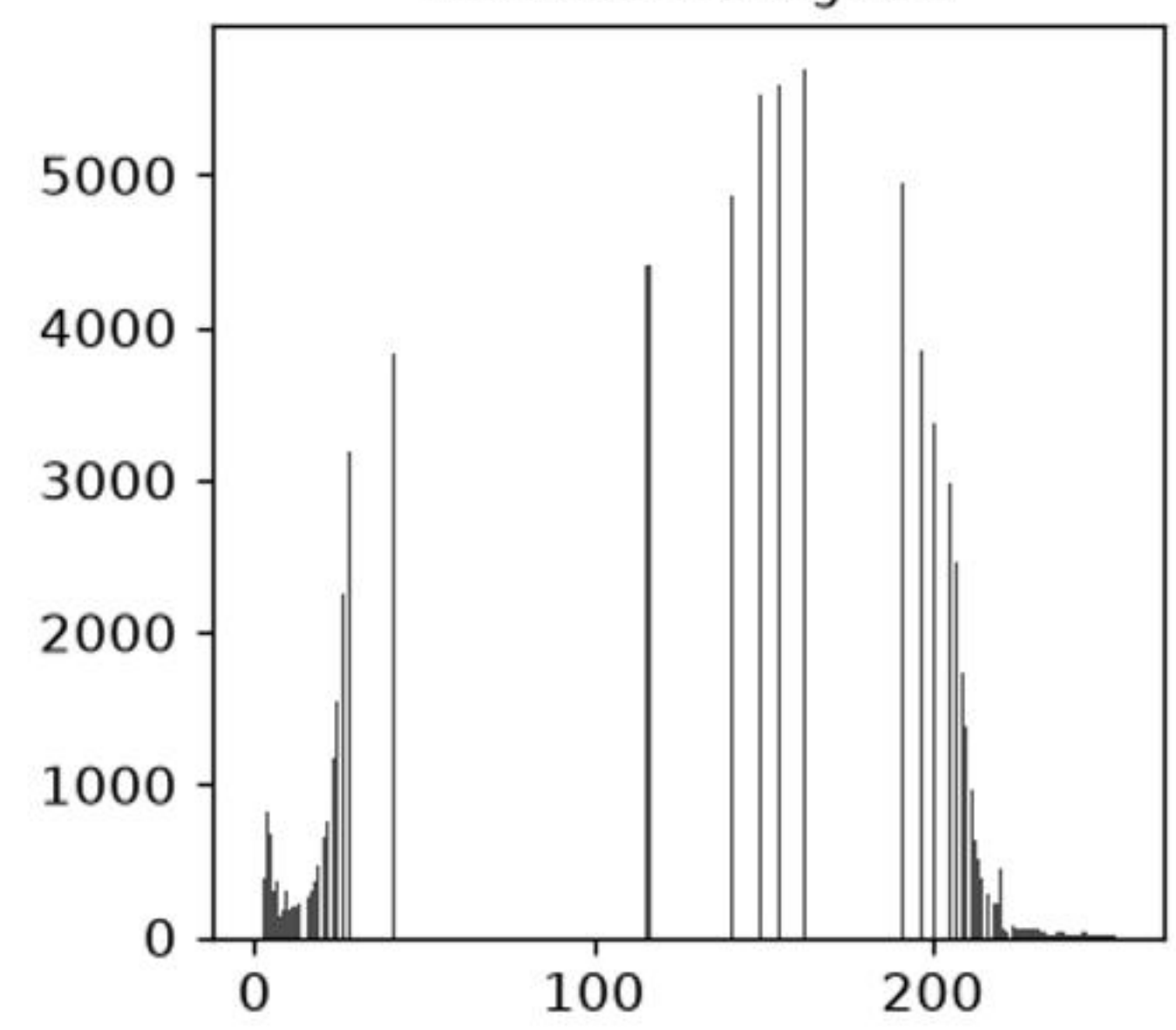
Equalized Histogram



Histogram Matched Image



Matched Histogram



Experiment 7 — Spatial Smoothing and Sharpening

Original



Smoothed (Mean Filter)

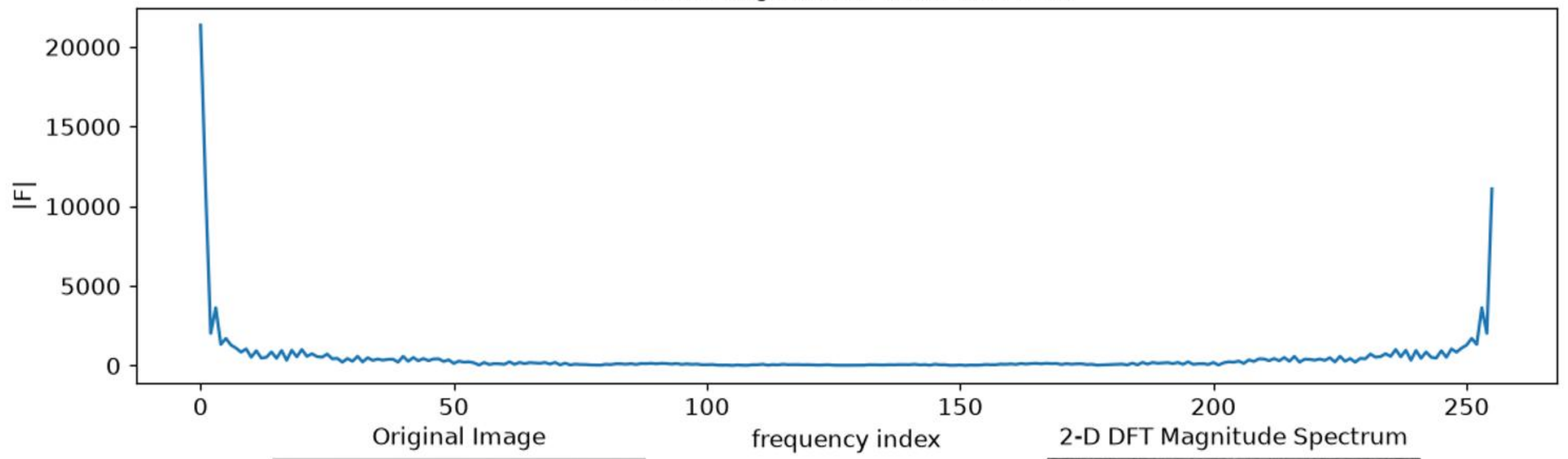


Sharpened (Laplacian)

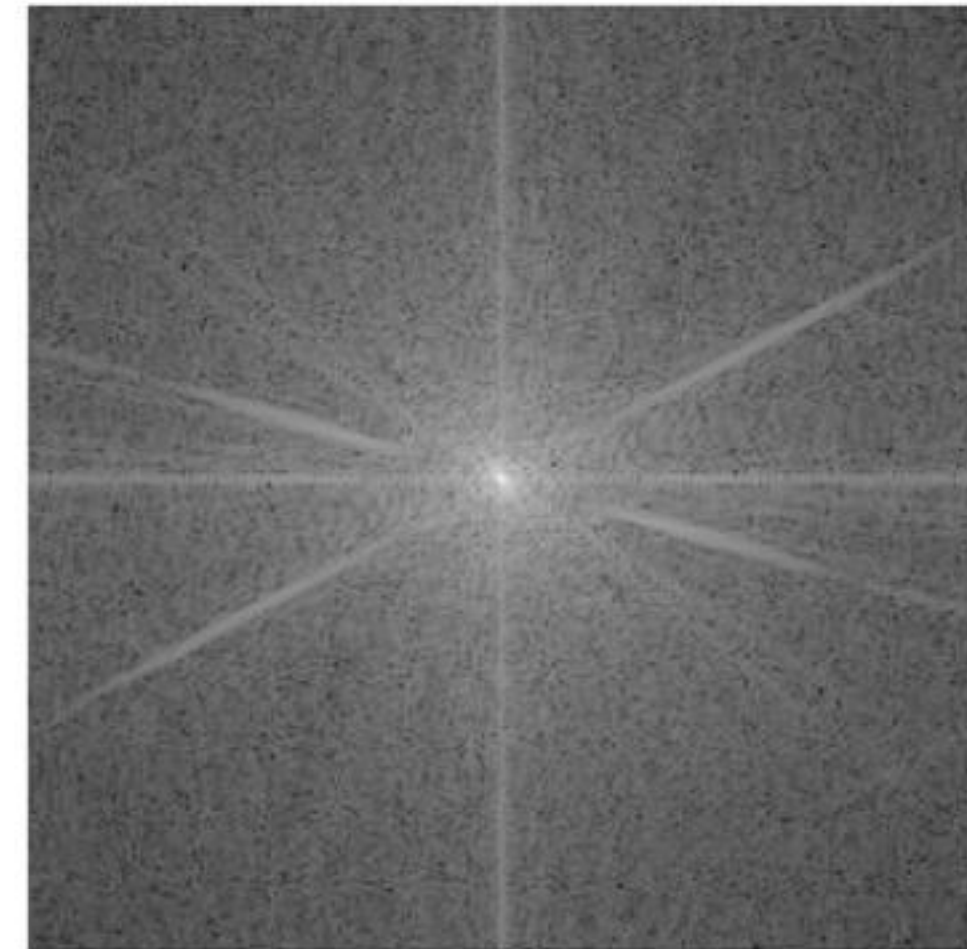


Experiment 8 — 1-D and 2-D Discrete Fourier Transform

1-D DFT Magnitude of a Row (row 128)



2-D DFT Magnitude Spectrum



Experiment 9 — Frequency-Domain Low-Pass and High-Pass Filtering

Original



Ideal LPF



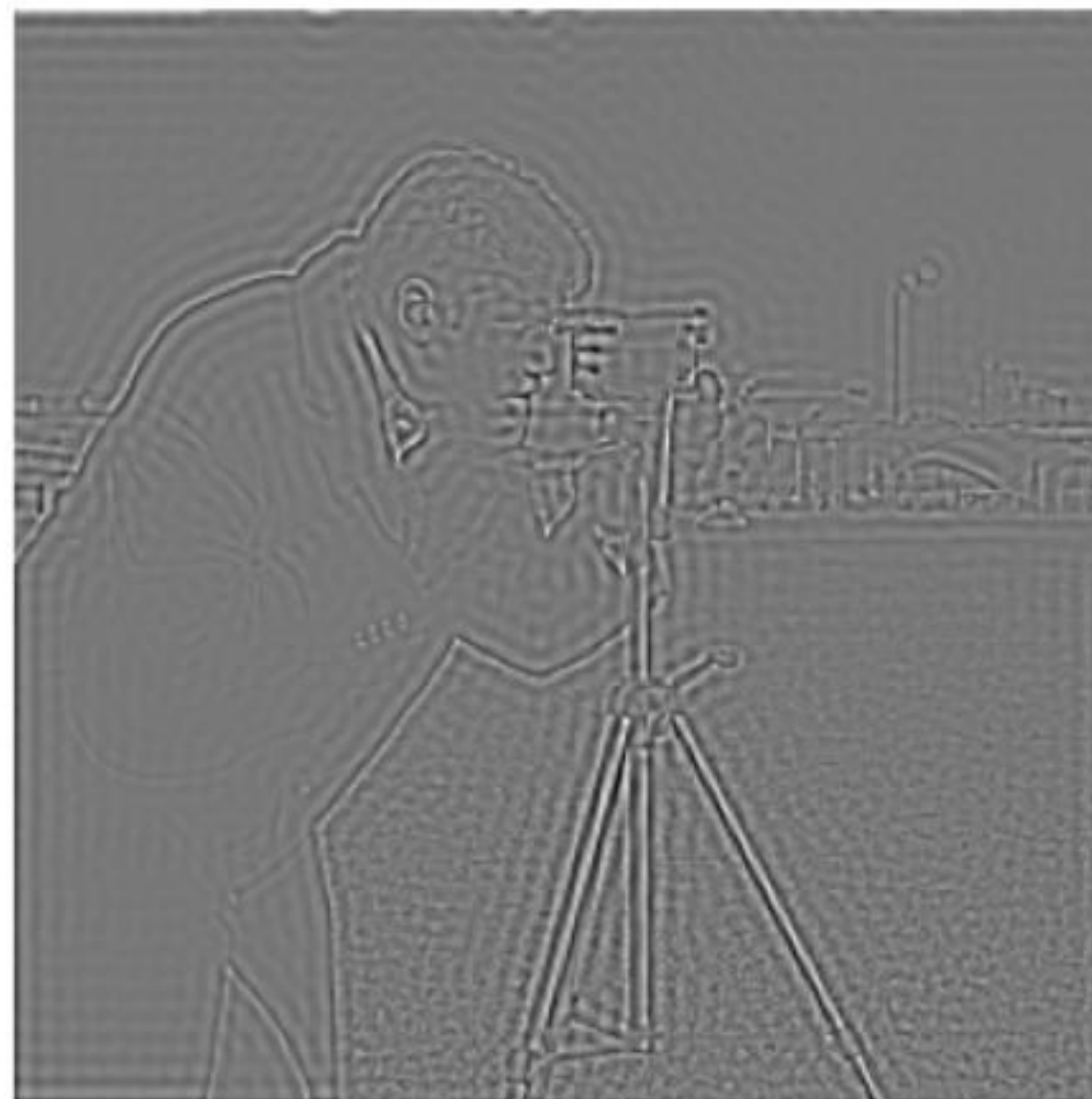
Butterworth LPF



Gaussian LPF



Ideal HPF



Gaussian HPF



Experiment 10 — Noise Modelling and Restoration

Original



Gaussian Noise



Mean Filter Restored



Original



Salt & Pepper Noise

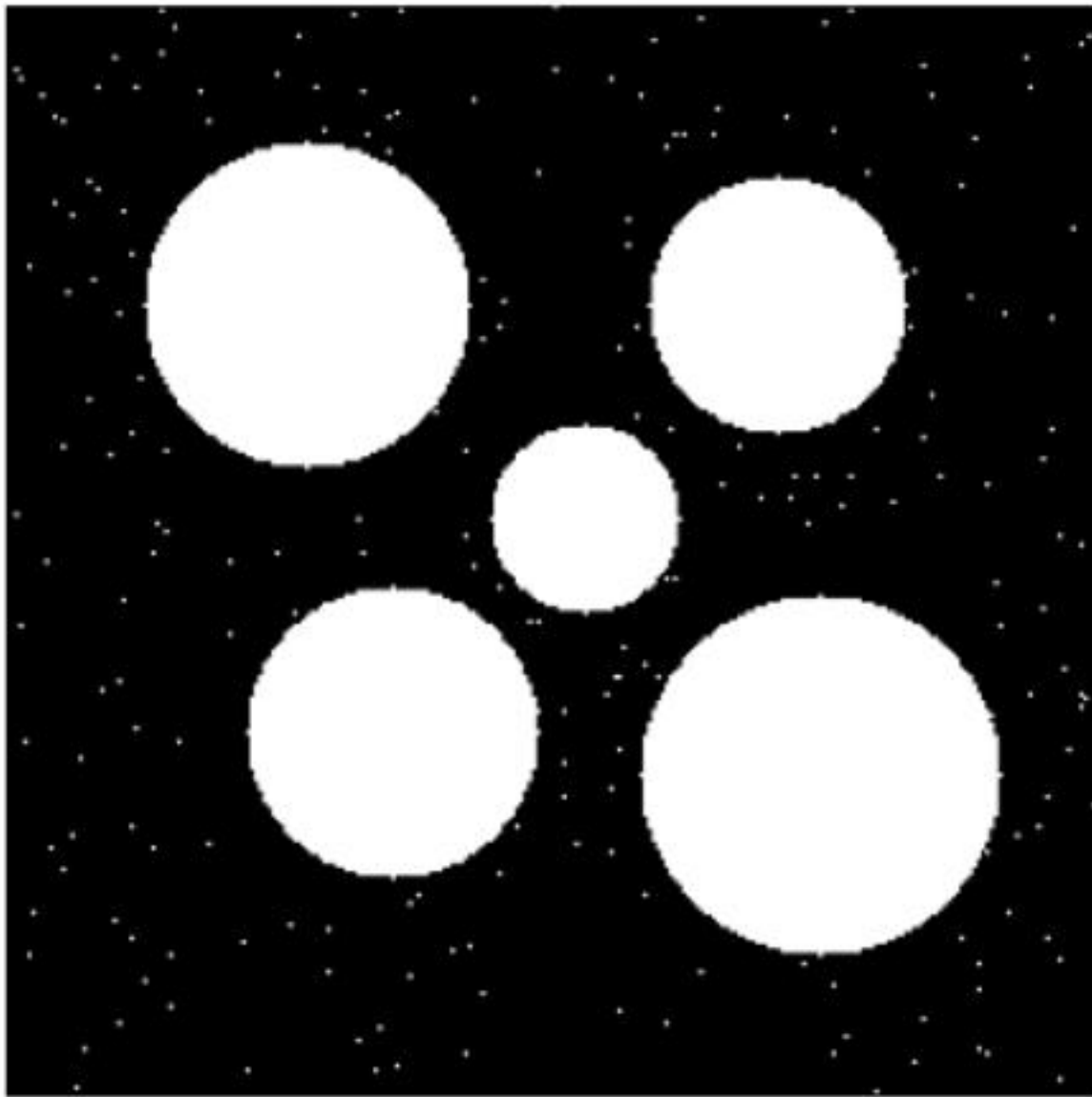


Adaptive Filter Restored

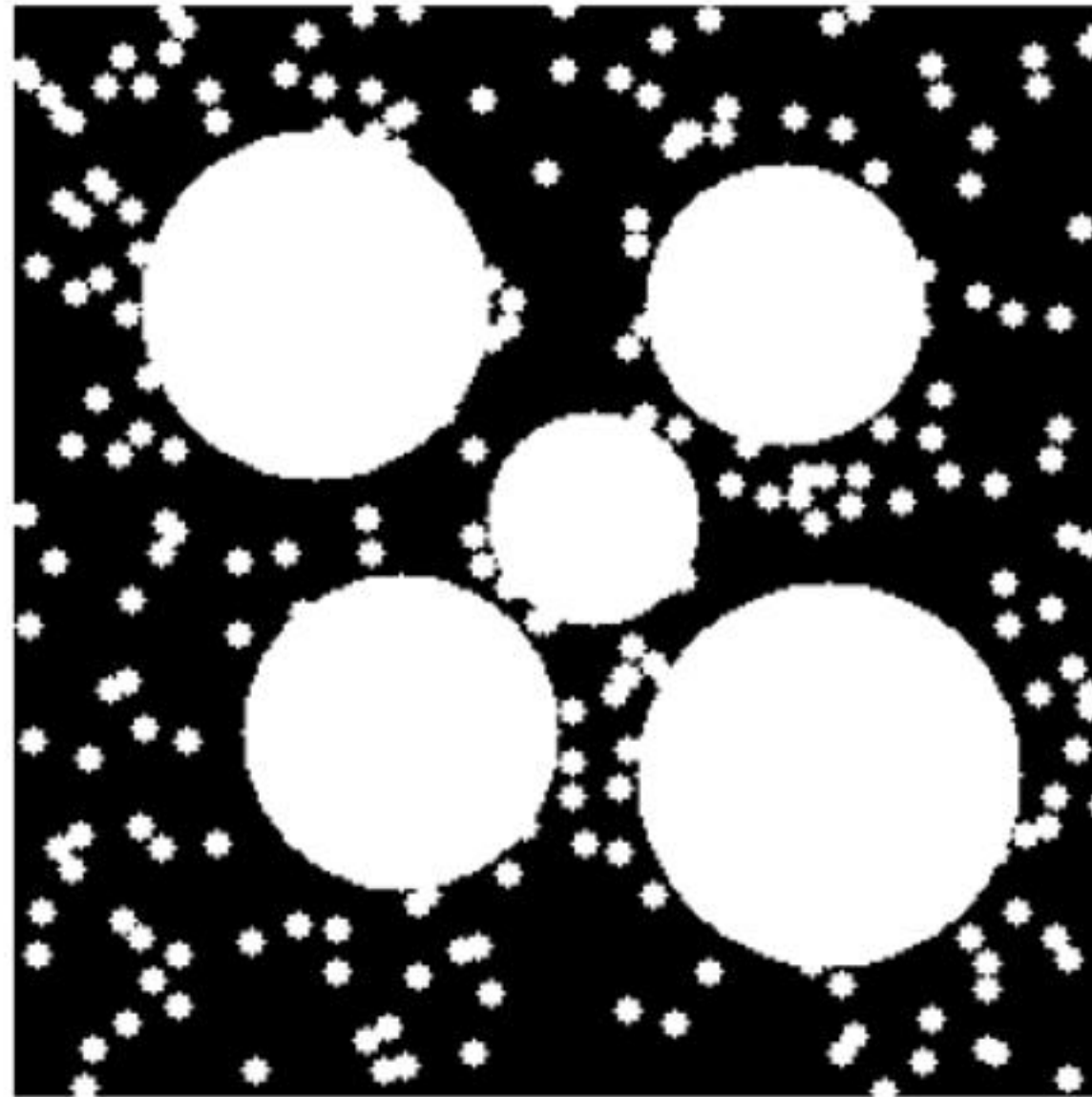


Experiment 11 — Morphological Operations

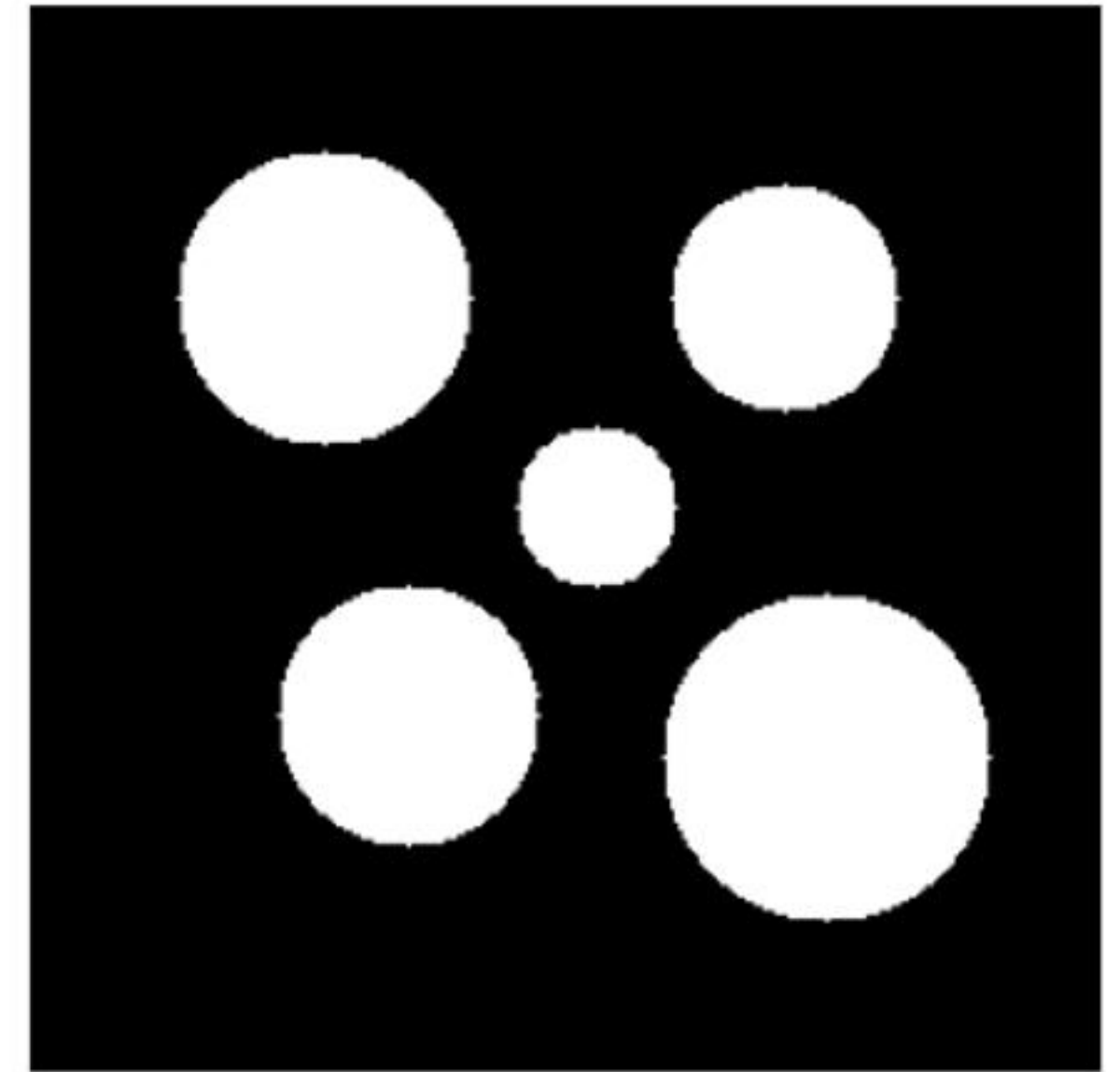
Original Binary Image



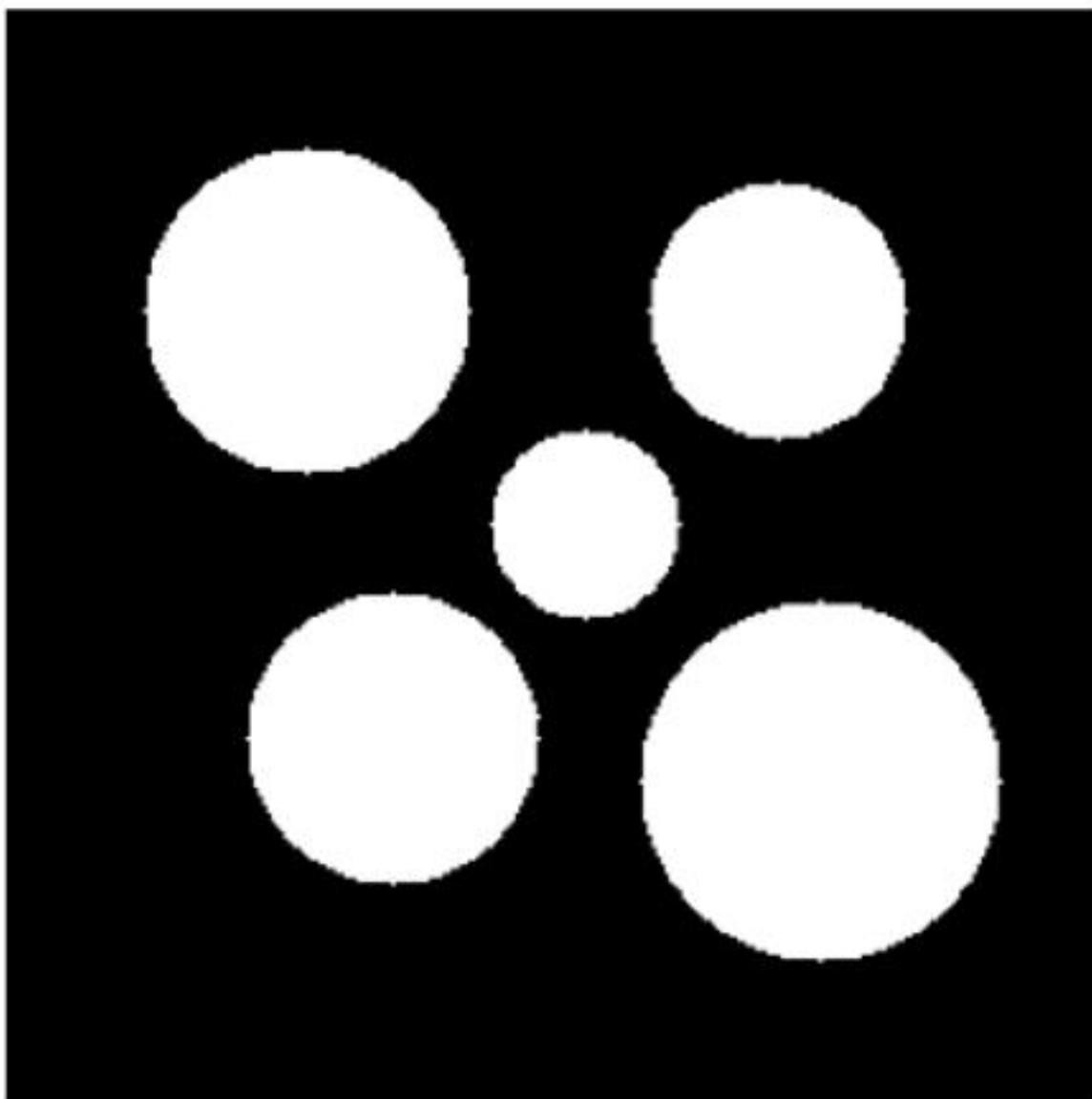
Dilation



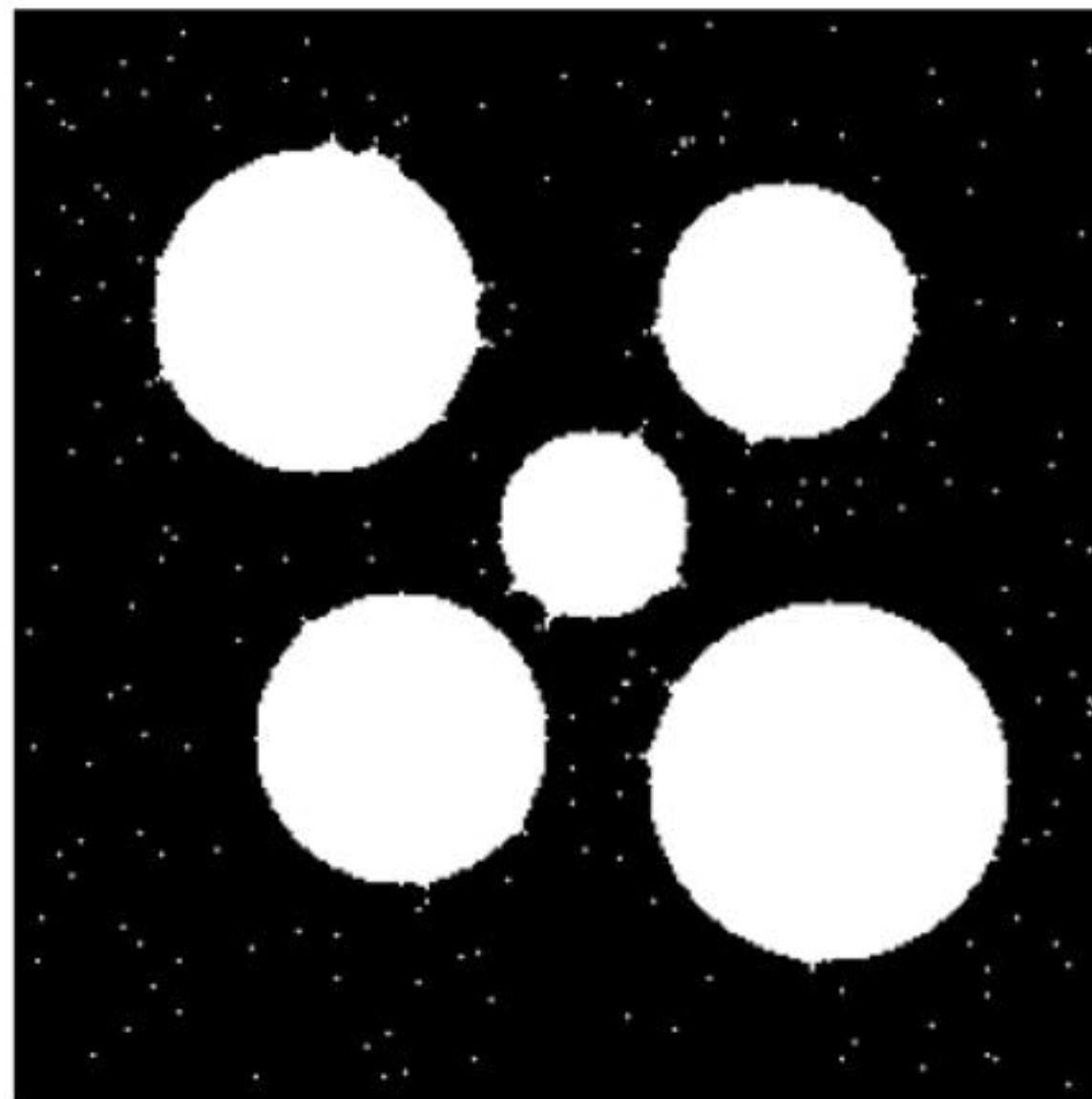
Erosion



Opening



Closing



Hit-or-Miss Result

